SECTION 1

Time—30 minutes

45 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. PRESERVE: (A) disagree (B) destroy (C) hide (D) distinguish (E) withdraw

2. FRUITFUL: (A) barren (B) unique (C) native (D) absorbent (E) extreme

3. COMPLIANCE: (A) sentimentality (B) vitality (C) sorrow (D) inexperience (E) rebellion

4. BYPASS: (A) enlarge (B) advance (C) copy (D) throw away (E) go through

5. DETERMINATE: (A) fundamental (B) luckless (C) inconclusive (D) highly selective (E) producing plentifully

6. ENHANCE: (A) detect (B) misplace (C) conquer (D) impair (E) distribute

7. COHERENT: (A) boring (B) disorganized (C) hostile (D) unrestrained (E) unfaithful

8. JOLLITY: (A) pallor (B) gravity (C) infamy (D) complexity (E) resourcefulness

9. FOIBLE: (A) offhand remark (B) strong point (C) freedom (D) accident (E) responsibility

10. EFFACE: (A) support (B) attack (C) deny (D) restore (E) suffer

11. PONDEROUS: (A) wet (B) shady (C) lightweight (D) mistaken (E) completed

12. DEMISE: (A) union (B) merit (C) birth (D) strain (E) bargain

13. ENIGMATIC: (A) relaxed (B) colorful (C) clear (D) contemptuous of authority (E) excluded from consideration

14. AMELIORATE: (A) agree to (B) make worse (C) disclaim (D) sequester (E) deviate

15. CAPRICIOUS: (A) listless (B) impartial (C) steadfast (D) truthful (E) malicious

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. Although the critics agreed that the book was brilliant, so few copies were sold that the work brought the author little ---- reward.

(A) theoretical (B) thoughtful (C) financial (D) abstract (E) informative

17. He is an unbeliever, but he is broad-minded enough to decline the mysteries of religion without ---- them.

(A) denouncing (B) understanding (C) praising (D) doubting (E) studying

18. Wellington, who reacted modestly to his fame, contrasted sharply with Marlborough, who ---- an ostentatious display of his glory.

(A) regretted (B) relished (C) disdained (D) minimized (E) discounted

19. Her ---- media coverage of Hispanic affairs in Los Angeles led Francisca Flores to publish a newsletter aimed at informing the Hispanic community of ---- events.

(A) complacency with. .popular
(B) unhappiness with. .uninteresting
(C) approval of. .unknown
(D) dissatisfaction with. .relevant
(E) confusion about. .imaginative

20. The phenomenon is called viral ---- because the presence of one kind of virus seems to inhibit infection by any other.

(A) proliferation (B) mutation (C) interference (D) epidemic (E) cooperation

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Princeton, N.J. 08541

1

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

"I always paint what I see," said Turner; yet no painter ever departed further from close imitation or took more liberties with subjects. He elongated steeples, reconstructed buildings, changed the course of rivers, even the course of the sun. Yet one critic says Turner was sincere in thinking that he painted what he saw; that he seemed to be "swept along as by a dream," and the changes he made "came into his head involuntarily."

But the ordinary person's idea of imitation is the setting down with the utmost precision every visible fact in a subject—"making it look like the real thing." Imitation in a painting is, for example, flowers that one can botanize, or fruit that "looks good enough to eat" and has a fly on it one feels the urge to brush off. Such deceptive resemblance gives people the pleasure of surprise rather than an aesthetic pleasure. Carefully realistic pictures may please, but the pictures people enjoy living with leave something to the imagination. "The secret of being a bore," said Voltaire, "is to tell everything."

Actually, few painters try to carry imitation to the point of deceptive resemblance. It cannot be done at all unless the object painted is small enough to be reproduced full size. A true artist never tries to make us believe that his or her medium is anything except a medium. It is paint with which great painters express themselves.

What the ordinary person generally means by imitation of a large subject—say a landscape—is a mirror copy much like a colored photograph: nature with its infinite mass of details crowded into the picture, unsifted, unassimilated, and unarranged. Even if it were possible to put in everything seen, it would not be art any more than copying a poem is creating poetry. The painter uses nature not as a copybook but as a source of inspiration, selects such details as suits his or her purpose, and arranges them in a picture with a discriminating regard for pictorial effect.

If a painting is an unsifted mass of objects, it fails at the very point where art begins, the point that marks the difference between art and imitation. One of the pathetic fallacies of art is that realism of fact creates realism of effect, that increasing the facts of nature in a picture or sculpture increases the "feeling of nature" in it. On the contrary, such techniques decrease the sensation because fullness of fact leaves too little to the imagination. That is the trouble with wax figures and with pictures in which every detail is perfectly defined. What could better give us the facts of nature than a stuffed bird with glass eyes? Yet an artist, employing only one percent of the facts of nature, can give us a rough sketch of a bird that is infinitely better.

21. According to the author, Turner treated nature very

(A) realistically and carefully
(B) crudely and indifferently
(C) dishonestly and carelessly
(D) imaginatively and freely
(E) sketchily and hastily

22. According to the passage, a true work of art can be the product of all the following EXCEPT

(A) an artist's inspiration
(B) the study of nature
(C) careful selection of details
(D) arrangement for a desired effect
(E) photographic realism

23. It can be inferred that the author quotes Voltaire (lines 20-21) to

(A) indicate that Voltaire was a discerning art critic
(B) suggest that people who do not know much about art are boring
(C) identify the source of good imagination
(D) confirm the need for a discriminating choice of details
(E) suggest that artists produce fewer paintings

24. It can be inferred from the passage that a painter who wished "to carry imitation to the point of deceptive resemblance" (lines 22-23) would find which of the following the easiest subject to attempt?

(A) A small vase of flowers
(B) A seascape
(C) A field with distant trees
(D) A street scene
(E) The interior of a small cottage

25. The author believes that an "ordinary" person (lines 10 and 29) would be most likely to agree with which of the following statements about art?

(A) Art and nature are far apart and must not be confused.
(B) A skillful taxidermist practices a craft, not an art.
(C) A photograph "imitates" nature far better than many paintings.
(D) If one recognizes everything in a painting, it is boring art.
(E) A photograph cannot surprise us and therefore cannot be art.

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

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Among the things that the Chinese achieved must be counted, of course, the three discoveries that Francis Bacon pointed out as unknown to his Western ancestors but instrumental in changing the face of the world: printing, gunpowder, and the magnet. (It is, of course, because of the early invention of printing in China that we can now know so much more of its antiquity than we do of that of the rest of the world.) Also specifically Chinese are the invention of the kite, with all the knowledge of practical aeronautics that accompanies such a development, and the invention of the seismograph, used to detect and locate earthquakes. There is still some uncertainty, however, as to the actual working mechanism of the latter. Less of a mystery, but just as ingenious, is the "south-pointing carriage," a device that seems to have used differential gearing to keep a figure pointing in a constant direction, no matter where the vehicle was led.

But let it not be thought that one is dealing only with mechanical inventions: the oracle-bones of the Shang period (ca. 1500 to 1000 B.C.) contain omens and astrological texts that match rather well those of the quite separate tradition represented by the clay tablets of the same period found in Mesopotamia, and the Chinese civilization had beautifully drawn maps with accurate coordinate systems far better in detail than those of the West. China also had the mathematics of Pascal's triangle for binomial expansions by A.D. 1303 and Horner's method for solving equations long before Horner. In China, mathematics never became the backbone of astronomical and scientific theory as it did in the West and, though there were transmissions in both directions, the two world cultures remained strikingly different in this important respect. It is, after all, this difference that led the West from Ptolemy to Kepler and Newton, and thence to Einstein. But the nonmathematical theories of chemistry and biology show China as a fair match for the best the West could attain and, in the techniques of industrialization, China scored huge successes in such areas as the production of cast iron and the sensible improvement of carriages and harnesses while Europe was still inventing feudalism and monastic contemplation.

One cannot say all these things, however, without blowing hot and cold with pride of ownership over who did what first and who got which idea from whom. We shall have to find out much more about both the European and the Chinese past before we can make any substantial estimate of the nature and importance of the transmissions between East and West.

26. The chief focus of the passage is on which of the following?

(A) Indicating the many similarities between Chinese and Western cultures
(B) Lamenting the decision of a brilliant culture to relinquish its leadership in the development of various sciences
(C) Showing that, unlike Chinese science, Western science is the result of individual achievements
(D) Proving that Chinese contributions to science were more significant than those of the West
(E) Describing some Chinese scientific achievements and comparing them to those of the West

27. The attitude toward Europe conveyed by the author's use of the words "still inventing" (line 45) is best described as one of

(A) delighted amazement
(B) mild condescension
(C) uncontrolled amusement
(D) grudging admiration
(E) growing anger

28. According to the passage, a divergence in Chinese and Western scientific traditions is most clearly indicated by

(A) Western use of mathematics in the development of scientific theory
(B) the early use of astrology among the Chinese
(C) the completely different mathematical discoveries of each culture
(D) the different mechanical inventions of each culture
(E) Chinese accomplishments with regard to industrial techniques

29. The author's attitude toward the Chinese achievements mentioned in lines 1-45 is best described as one of

(A) disbelief (B) admiration (C) anxiety
(D) ambivalence (E) apathy

30. The final statement in the passage chiefly suggests which of the following?

(A) The history of science is worth study in almost any culture.
(B) Chinese scientific achievement is such that the West would profit greatly from studying it.
(C) More investigation is needed to determine exactly what influence China and the West had on each other.
(D) Two attitudes toward science as divergent as those of China and the West will be hard to reconcile.
(E) Neither China nor the West will ever admit what each has learned from the other.

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1

Select the word or set of words that best completes each of the following sentences.

31. This ---- painter was but ---- pianist; yet she seemed more pleased by compliments on her playing than on her painting.

(A) amateur. .a professional
(B) unexceptional. .an untutored
(C) accomplished. .a preeminent
(D) virtuoso. .a mediocre
(E) novice. .a renowned

32. In comparative anatomy, all methods of analysis, save direct observation, are beset with ---- which require great ---- if reliable conclusions are to be drawn.

(A) analogies. .duplication
(B) problems. .supposition
(C) premises. .inducement
(D) pitfalls. .circumspection
(E) judgments. .prerogative

33. The primary quality that a leader must possess is a single-minded dedication to the people's welfare, whereby ---- interests are ---- to the people's interests.

(A) partisan. .subordinated
(B) special. .ascribed
(C) conflicting. .attributed
(D) factional. .preferred
(E) dictatorial. .relegated

34. My heretofore faithful friends, considering their ---- hostile actions against me, must now, ----, be numbered among my enemies.

(A) resentment of. .justifiably
(B) discouragement of. .indiscriminately
(C) revival of. .simultaneously
(D) complicity in. .lamentably
(E) acquiescence in. .auspiciously

35. The candidate's writers showed poor judgment in embroidering the speech with that apocryphal anecdote; it was certainly not an appropriate ----, and thus was as ---- as it was artificial.

(A) development. .pivotal
(B) proposal. .astounding
(C) novelty. .insufficient
(D) digression. .gratuitous
(E) improvisation. .expedient

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. DANCER : FEET :: (A) surgeon : heart
(B) juggler : hands (C) drummer : drums
(D) conductor : voice (E) musician : eyes

37. RACKET : TENNIS :: (A) puck : hockey
(B) rifle : duck (C) hammer : nail
(D) ball : soccer (E) bat : baseball

38. FERTILIZER : PLANT :: (A) pollen : allergy
(B) bandage : wound (C) flame : fire
(D) nitrogen : atmosphere (E) food : human

39. LIE : DECEIT :: (A) cunning : guilt
(B) wealth : generosity (C) suspicion : distrust
(D) punishment : error (E) loyalty : treason

40. PAPYRUS : PAPER :: (A) canoe : kayak
(B) scroll : book (C) magazine : rack
(D) cave : cavern (E) parchment : bark

41. ISLAND : ARCHIPELAGO :: (A) castle : moat
(B) star : galaxy (C) river : delta
(D) bay : peninsula (E) earth : hemisphere

42. PHOENIX : IMMORTALITY ::
(A) unicorn : cowardice (B) sphinx : mystery
(C) salamander : speed (D) ogre : wisdom
(E) chimera : stability

43. STIPEND : BONUS :: (A) performance : encore
(B) engagement : marriage (C) gift : celebration
(D) reward : superiority (E) battle : victory

44. MATRIARCHY : WOMAN :: (A) legality : lawyer
(B) monarchy : sovereign (C) hierarchy : heir
(D) feudalism : farmer (E) duplicity : thief

45. PERFUNCTORY : ENTHUSIASM ::
(A) hostile : animosity
(B) submissive : defiance
(C) flagrant : criticism
(D) solitary : conviction
(E) honorary : admiration

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 2

Time—30 minutes

25 QUESTIONS

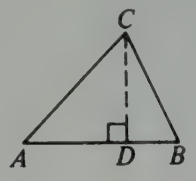
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. A number 1,010 more than 100,101 is

   (A) 100,111 (B) 100,211 (C) 101,101
   (D) 101,111 (E) 111,111

2. If $\dfrac{x}{y} = 3$ and $x = 12$, then $x - y =$

   (A) 3 (B) 5 (C) 6 (D) 8 (E) 9

3. If a, b, c, d, and e are whole numbers, the expression $a(b(c + d) + e)$ will be an even number whenever which of the following is even?

   (A) a (B) b (C) c (D) d (E) e

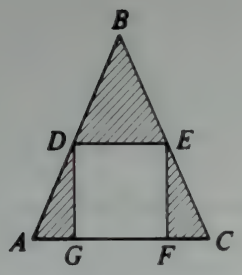
EMPLOYMENT PATTERN IN COMPANY X

Year	1	2	3	4	5
Number of Workers	23	69	207	621	

4. The number of workers employed by a company has increased by a constant multiple from year to year as shown in the chart above. If this pattern continues, how many workers can the company expect to employ in year 5 ?

   (A) 835
   (B) 1,035
   (C) 1,242
   (D) 1,463
   (E) 1,863

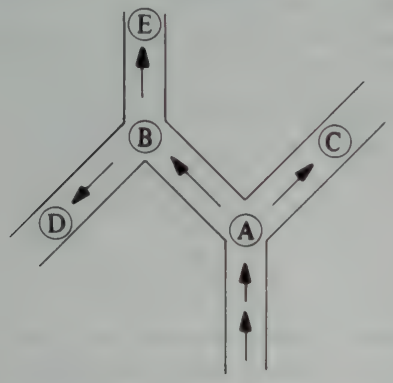
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5. In the figure above, the area of triangle ABC is 32 and the total area of the shaded regions is 16. If $DEFG$ is a square, what is the length of segment GF?

(A) 4 (B) 5 (C) $\sqrt{34}$ (D) 6 (E) $5\sqrt{2}$

Questions 6-7 refer to the information below.



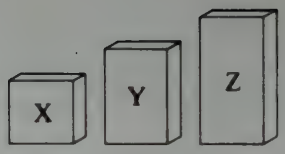
A traffic survey shows that of every 12 cars that arrive at point A, 7 turn toward point B and 5 turn toward point C. Of every 7 cars that arrive at point B, 3 turn toward point D and 4 turn toward point E.

6. Of 48 cars arriving at point A, how many are expected to turn toward point B?

(A) 15 (B) 20 (C) 28 (D) 30 (E) 36

7. Of 72 cars arriving at point A, how many are expected eventually to turn toward point D?

(A) 18 (B) 24 (C) 26 (D) 32 (E) 42



8. In the figure above, the number of grams of cereal in box Y is greater than the number in box X and less than the number in box Z. If box Z contains 500 grams, each of the following could be the total number of grams of cereal in the three boxes EXCEPT

(A) 900 g
(B) 1,000 g
(C) 1,200 g
(D) 1,350 g
(E) 1,500 g

---

9. The sum of the digits of a three-digit number is 12. If the hundreds' digit is 3 times the tens' digit and the tens' digit is $\frac{1}{2}$ the units' digit, what is the number?

(A) 312
(B) 624
(C) 912
(D) 936
(E) 963

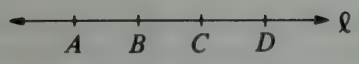
10. The average (arithmetic mean) mass of 4 objects is 50 kilograms. If 3 of the objects each have a mass of 45 kilograms, what is the mass, in kilograms, of the fourth object?

(A) 70 (B) 65 (C) 60 (D) 55 (E) 50

11. If 10 is 5 percent of N, then $N =$

(A) 2
(B) 5
(C) 20
(D) 50
(E) 200

GO ON TO THE NEXT PAGE



12. Which of the following statements is (are) true of the lengths of the segments on line ℓ above?

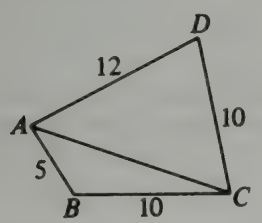
I. $AB + CD = AD$
II. $AB + BC = AD - CD$
III. $AC - AB = AD - CD$

(A) I only (B) II only (C) III only
(D) I and II only (E) I, II, and III

---

13. If $Q = \frac{(x+y)h}{2}$, what is the average (arithmetic mean) of x and y when $Q = 20$ and $h = 5$?

(A) 100
(B) $\frac{25}{2}$
(C) 10
(D) 4
(E) 2



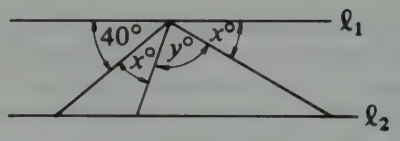
14. In the figure above, the perimeter of $\triangle ACD$ is how much greater than the perimeter of $\triangle ACB$?

(A) 22 (B) 14 (C) $5\sqrt{5}$ (D) 7 (E) $\sqrt{2}$

---

15. $\frac{4.5x^6y}{0.3x^2y^2} =$

(A) $\frac{0.15x^4}{y^2}$
(B) $\frac{1.5x^4}{y}$
(C) $\frac{1.5x^3}{y^2}$
(D) $\frac{15x^4}{y}$
(E) $\frac{15x^3}{y}$



16. In the figure above, if lines ℓ_1 and ℓ_2 are parallel, what is the value of y?

(A) 30
(B) 40
(C) 70
(D) 80
(E) It cannot be determined from the information given.

---

17. This question was not counted in computing scores.

---

18. If half the people in a room leave at the end of every five-minute interval and at the end of twenty minutes the next to the last person leaves, how many people were in the room to start with? (Assume that no one enters the room once the process begins.)

(A) 32
(B) 28
(C) 16
(D) 12
(E) 8

GO ON TO THE NEXT PAGE

19. Susan finds approximately f four-leaf clovers in every square meter of a field. If the field is a rectangle l meters long and w meters wide, what is the best approximation of the number of four-leaf clovers she finds?

(A) flw

(B) fl^2w^2

(C) $\frac{f^2}{lw}$

(D) $\frac{f}{lw}$

(E) $\frac{lw}{f}$

20. What fraction of the number of integer multiples of 2 between 1 and 99 are also integer multiples of 3 ?

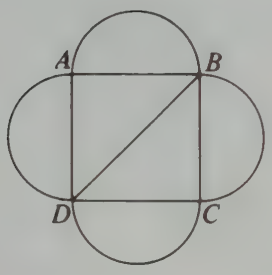
(A) $\frac{16}{99}$

(B) $\frac{16}{49}$

(C) $\frac{1}{3}$

(D) $\frac{49}{99}$

(E) $\frac{33}{49}$



21. In the figure above, $ABCD$ is a square and each arc shown is a semicircle. If diagonal BD is equal to $\sqrt{2}$, what is the sum of the areas of the four semicircles?

(A) $\frac{\pi}{2}$ (B) π (C) $\sqrt{2}\pi$

(D) 2π (E) $2\sqrt{2}\pi$

22. Seven cards in a pile are numbered 1 through 7. One card is drawn. The units' digit of the sum of the numbers on the remaining cards is 7. What is the number on the drawn card?

(A) 1
(B) 3
(C) 5
(D) 7
(E) It cannot be determined from the information given.

23. If for any number x, $\boxed{x}$ is defined as the least integer that is greater than or equal to x, then $\boxed{-2.5} + \boxed{12} =$

(A) 9 (B) 9.5 (C) 10 (D) 11 (E) 14.5

24. A 25-foot ladder is placed against a vertical wall of a building, with the bottom of the ladder standing on concrete 7 feet from the base of the building. If the top of the ladder slips down 4 feet, then the bottom of the ladder will slide out

(A) 4 ft
(B) 5 ft
(C) 6 ft
(D) 7 ft
(E) 8 ft

25. The length of rectangle S is 20 percent longer than the length of rectangle R, and the width of rectangle S is 20 percent shorter than the width of rectangle R. The area of rectangle S is

(A) 20% greater than the area of rectangle R
(B) 4% greater than the area of rectangle R
(C) equal to the area of rectangle R
(D) 4% less than the area of rectangle R
(E) 20% less than the area of rectangle R

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 3

Time—30 minutes

40 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. GUSTY: (A) warm (B) calm (C) bright (D) young (E) thin

2. RETREAT: (A) wander (B) budge (C) advance (D) travel (E) stumble

3. PREMEDITATED: (A) unconventional (B) undeliberated (C) unfathomable (D) independent (E) insubordinate

4. IOTA: (A) termination (B) primary cause (C) unmixed solution (D) inconsistency (E) large quantity

5. ABSTRUSE: (A) pure (B) colorless (C) lengthy (D) readily understood (E) cordially welcomed

6. CODDLE: (A) summon (B) clarify (C) abuse (D) separate (E) confess

7. IRREPROACHABLE: (A) isolated (B) essential (C) adjustable (D) recognizable (E) reprehensible

8. RAREFY: (A) obstruct (B) manipulate (C) make denser (D) use carelessly (E) perform faithfully

9. FETTER: (A) interrogate (B) implicate (C) placate (D) decorate (E) liberate

10. NEOLOGISM:
(A) nameless article
(B) foreign object
(C) exaggerated movement
(D) impoverished condition
(E) obsolete expression

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. If oil and coal are constantly ----, then someday the natural supply will be ----.
(A) produced. .available
(B) sought. .discovered
(C) supplanted. .reduced
(D) consumed. .exhausted
(E) burned. .useless

12. The 1970's witnessed an unprecedented ---- of the scope of historical studies; the result was that the distinctive experiences of Black Americans, Native Americans, women, and other previously ---- groups moved to the forefront of inquiry.
(A) redefinition. .neglected
(B) recapitulation. .uninvestigated
(C) summarizing. .well-recognized
(D) illumination. .significant
(E) elaboration. .chronicled

13. Although the substance is normally quite ----, scientists found that when tempered with other elements it could be stored safely in metal containers.
(A) voluminous (B) caustic (C) insoluble (D) vapid (E) heterogeneous

14. Norton is a drama critic who is evidently never ----; he writes a ---- review of every play he sees.
(A) astonished. .dazed (B) impressed. .rave (C) disgruntled. .peevish (D) pleased. .carping (E) satisfied. .reverential

15. Games and athletic contests are so intimately a part of life that they are valuable ---- the intensity and vitality of a given culture at any one time.
(A) modifiers of (B) antidotes to (C) exceptions to (D) obstacles to (E) indices of

GO ON TO THE NEXT PAGE

3

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. STABLE : HORSE :: (A) kennel : dog
(B) trap : fox (C) flower : bee
(D) hutch : cat (E) forest : bird

17. DEGREE : TEMPERATURE :: (A) heat : cold
(B) activity : pace (C) bank : currency
(D) minute : time (E) acre : volume

18. BOW : CELLO :: (A) mute : trumpet
(B) pedal : piano (C) string : violin
(D) skin : drum (E) pick : banjo

19. COWER : FEAR :: (A) pursue : despair
(B) wobble : power (C) plod : energy
(D) rove : speed (E) strut : pride

20. CROWD : NUMBER :: (A) room : wall
(B) table : leg (C) street : sidewalk
(D) treasure : worth (E) painting : variety

21. BUNGLER : SKILL :: (A) fool : amusement
(B) critic : error (C) daredevil : caution
(D) braggart : confidence (E) genius : intelligence

22. CONVOKE : ASSEMBLY :: (A) muster : troops
(B) challenge : opponents (C) veto : legislation
(D) canvass : membership (E) disband : army

23. ANECDOTE : NARRATIVE :: (A) dogma : belief
(B) vendetta : revenge (C) code : signal
(D) snack : meal (E) jester : comedy

24. ASSUAGE : ANGUISH ::
(A) investigate : annoyance (B) discuss : torment
(C) alleviate : affliction (D) assert : hatred
(E) inflame : pain

25. ALCHEMIST : TRANSMUTATION ::
(A) traitor : loyalty
(B) proselytizer : conversion
(C) scientist : equivocation
(D) anarchist : tranquillity
(E) astronomer : consternation

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

The warden slowly turned his glass. Then he said, "I will now present you with proof of a total inner security. I could answer, 'I was director of a theater' or 'the leading actor in a repertory company.' In actual fact, I was the reviewer on a pretentious literary weekly. In other words, an intellectual dilettante." The warden sipped his whisky. "You see? I no longer feel the need to gild the dandelion. I have reached the degree of detachment and security where I can be ruthless without rancor, even toward myself." He pierced the yolk of a fried egg with his fork. "And you, Doctor? Of which illustrious medical college were you the youthful Chief Internist?"

The doctor hesitated; then the whisky seduced him. "I worked for a rural general practitioner just before I came here."

"Must have been fascinating."

"Disgusting. I spent most of my time being ashamed of myself."

"It is the privilege of youth to spend most of its time being ashamed of itself. You have progressed since then, I trust?"

"Somewhat."

The warden gave him a searching look. "Don't tell me you still indulge in attacks of conscience? That's a luxury you can ill afford in a place like this." He poured more whisky for them. "Well, here's to us: two forlorn nonentities on the pinnacle of power in a forgotten place. Cheers."

They drank.

The doctor looked at him uneasily. This line of conversation held a hidden menace, something he could not define but sensed quite clearly.

26. It can be inferred from the passage that the warden considers himself well qualified for his present job because of his ability to

(A) feign servility to his superiors
(B) remain emotionally aloof
(C) devote unusually long hours to his work
(D) require little in the way of physical comforts
(E) supplement his career with active participation in the arts

27. The warden's toast in lines 28-30 LEAST expresses which of the following?

(A) Sarcasm (B) Self-pity (C) Bitterness (D) Mistrust (E) Resignation

28. It can be inferred that the doctor's uneasiness at the end of the passage (lines 32-34) serves to forecast

(A) a sudden increase in the number of prisoners
(B) an abrupt shift in command at the prison
(C) the possibility of conflict between himself and the warden
(D) the warden's increasing inability to remain objective about his work
(E) the imminent arrival of another doctor in the prison

Our Sun was born, leading astronomers now believe, about five billion years ago, the spawn of a gigantic swirling nebula of gas and dust. As the particles coalesced, gravitation pulled them swiftly together and they heated up. Within a few million years the new body developed an interior temperature of millions of degrees, enough to ignite the processes of nuclear fusion. Hydrogen atoms fused to form helium atoms, giving off energy in the process, and the Sun began to shine.

The Sun has changed little since then, and likely will change little until the hydrogen at its center nears exhaustion, in perhaps another five billion years. Then, according to well-developed theories of stellar evolution, the outer layers will begin to expand and the Sun will swell and redden. Eventually it will loom across 25 percent of the Earth's sky, having increased a hundred times in diameter and a thousand times in brightness. Its hot breath may melt Mercury and Venus. The Earth may be reduced to a desolate, baking rock, its oceans completely boiled away. And earthlings, if they have survived, will have fled to some more hospitable part of the galaxy.

For a hundred million more years the swollen Sun will continue to deplete its nuclear fuel until finally only a tiny core will remain, and a frozen, presumably lifeless Earth will swing bleakly around a faint Sun no larger in the sky than the tiny planet Mars.

29. The primary purpose of the passage is to

(A) describe the relation between the Sun and the rest of the galaxy
(B) discuss a theory about the formation of the solar system
(C) give one example of the importance of the Sun to our universe
(D) sketch out the theoretical life cycle of the Sun
(E) explain the processes of nuclear fusion

30. Which of the following does the author most clearly imply about the sequence of events outlined in the passage?

(A) It is completely certain.
(B) It is possible, but rather unlikely.
(C) It is highly improbable.
(D) It represents one of many theories on the subject.
(E) It represents the best current thinking on the subject.

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31. It can be inferred from the passage that all of the following might explain why the author describes the Earth as "presumably lifeless" (line 28) EXCEPT:

(A) Surviving earthlings will probably have fled to a more hospitable climate.
(B) If the Earth is frozen, it will probably also be lifeless.
(C) Although the Earth will probably be frozen, the climate might become warmer again.
(D) Although human life will no doubt vanish from Earth, some other forms might still exist.
(E) The heat of the Sun will probably be insufficient to allow much life as we know it.

Virginia Woolf, writing from the vantage point of twentieth-century England, once speculated on the fate of a woman born in Shakespeare's day with a touch of Shakespeare's genius: she imagined a life marred by family tyranny, social hostility, thwarted ambition, and, finally, suicide. But had Woolf ever heard of Anne Bradstreet (who was born in Shakespeare's forty-eighth year), she would have had to draw up an entirely different scenario that would involve emigration to the New World. For Bradstreet had an excellent education and received from her society, the middle-class Puritan elite, every encouragement to write and publish. Although she was married at sixteen and raised eight children—conditions that Virginia Woolf thought inimical to literary composition—Bradstreet became the first professional woman poet writing in the English language after—and surely because—she left England at eighteen and came to America.

Yet if Anne Bradstreet is remembered today in America, it is not, correctly, as the American colonies' first poet, but as a "Puritan poet," even though her 1650 edition is an essentially secular volume—not only not "Puritan," but not particularly Christian. She did write important religious verse, but at a later time. However, we have no warrant for describing her as unrepresentative, for it was the New England Puritans who insisted that she be published, and who gave Bradstreet's first book the memorable title, with its boasting reference to both the sexual and national identity of the poet, *The Tenth Muse Lately Sprung Up in America*.

32. According to the passage, Virginia Woolf's scenario did not take into account Anne Bradstreet's success because

(A) Anne Bradstreet was not a contemporary of Shakespeare
(B) Anne Bradstreet was not primarily known as a poet
(C) Virginia Woolf did not know about Anne Bradstreet
(D) Virginia Woolf did not like Anne Bradstreet's poetry
(E) Virginia Woolf had limited her consideration to women born in England

33. The author apparently believes that Anne Bradstreet can be correctly described as a "Puritan poet" (line 22) because Anne Bradstreet

(A) lived in a Puritan society that encouraged her career
(B) acted as an apologist for the Puritans, explaining their views
(C) received most of her education and training from the American Puritans
(D) chose only religious subjects of interest to New England Puritans
(E) wrote poetry that could have appeared only in Puritan New England

34. The author suggests that the title of Anne Bradstreet's first book is significant primarily because it

(A) emphasized the Puritans' control over publications in their colony
(B) omitted her name and the content of the work
(C) acknowledged her as America's first English-speaking poet
(D) defied the existing traditions associated with Puritan poetry
(E) reflected her society's esteem for her as a poet, a woman, and an American

35. The author apparently believes that the most significant factor leading to Anne Bradstreet's success as a poet was her

(A) piety
(B) education
(C) early marriage
(D) emigration to America
(E) association with other writers

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In the medical sciences the word "normal" is in constant use but, as a rule, without proper clarification of its meaning. It is commonly held to be synonymous with "healthy," but health too awaits a better definition and a closer study. It is an old and, to a large extent, just criticism of physicians that they have considered the symptoms, causes, and measurements of disease without sufficient reference to the symptoms, causes, and measurements of health; that they have refined their methods of detecting departures from the normal without first reviewing the idea of normal or discussing its limitations. That "normal," in biological usage, means something other than "not deviating from a fixed standard" can scarcely be disputed: in all animals, variation is so constantly at work that no rigid pattern—whether anatomical, physiological, psychological, or immunological—is possible. Yet, due to the neglect of the study of "healthy" populations and of what may best be described as "normal variability," many findings have been judged abnormal (in the sense of pathological) with quite inadequate justification.

When we speak of normal variability, we must distinguish between two types: individual variability across time and under differing conditions of activity; and variability under set conditions of age, sex, and activity in a particular population or species. In the individual (apart from the slower changes of growth, maturity, and senescence) functional fluctuations—for instance, of pulse rate, blood pressure, body temperature, and tissue chemistry—occur from minute to minute. Within a particular population or species, we find those variations that account for differences among individuals, even under standard conditions, in their healthiest moods and most favored communities—variations that help to explain the differing resistances and susceptibilities of individuals to stress and other harmful agencies.

A moment's thought will suggest that both types of variability are essential to survival. Without continuous and swift physiochemical variations in the individual, there could be no adaptation to constantly changing environmental conditions, or to the impositions of action and inaction or fasting and repletion, or to the ebb and flow of emotional stimuli. Without species variations, there could be no communal equilibrium or adaptability, no maintenance of a stock by the emergence of types better fitted for survival in a changing and exacting world, no natural distribution of functions within the group, no slow adaptations to changes in the external environment and, in the case of humans, no scope for adjustment to changing occupations and social systems. In brief—in both the individual and the community—there could be no adaptability without variability. Complete standardization is incompatible with life.

36. Which of the following best restates the criticism of physicians mentioned in lines 5-13?

(A) Physicians sometimes fail to consider adequately the symptoms, causes, and measurements of disease.
(B) Physicians may decide to bring up to date the old definitions of "normality" and "health."
(C) Physicians do not regularly diagnose a departure from the normal as a pathological abnormality.
(D) Physicians have devoted more time and effort to the treatment of ill health than they have to the study of what is meant by "healthy."
(E) Physicians do not define illness in sufficient detail to be able always to recognize and study deviations from the normal.

37. In the second paragraph, the author's chief distinction is between which of the following?

(A) Slow changes and reversible changes
(B) Normal variability and abnormal variability
(C) Temporal variability and variations in heredity
(D) Changes within a single person and differences within a group
(E) Functional fluctuations and standard biological conditions

38. According to the passage, individual variability is directly responsible for all of the following EXCEPT

(A) dealing with human emotions
(B) adjusting to nutritional fluctuations
(C) contending with impermanent surroundings
(D) coping with changing levels of activity
(E) maintaining communal equilibrium

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39. The author concludes that "complete standardization is incompatible with life" (lines 58-59) primarily because

(A) the great number of individuals prevents the establishment of patterns of normality
(B) there has been a neglect of the study of healthy populations and of normal variability
(C) a significant level of variation must be in operation to ensure survival
(D) functional fluctuations are directly responsible for the ability of a species to adapt over time
(E) humans would take longer to adjust to changing social conditions without variation

40. The primary purpose of the passage is to

(A) criticize physicians for their apparent disregard of the symptoms, causes, and measurements of normal variability
(B) describe the differences between "normal" and "abnormal" in the biological sense of the words
(C) elaborate the idea of a fixed standard of health and normality
(D) advocate the concept and clarify the meaning of "normal variability"
(E) differentiate among various types of conditions usually labeled "normal"

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 5

Time—30 minutes

35 QUESTIONS

In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

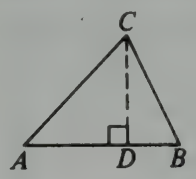
Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$

The number of degrees of arc in a circle is 360.

The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $2x + 1 = 5$, then $4x =$

   (A) 8 (B) 10 (C) 12 (D) 16 (E) 24

2. A florist buys roses at \$0.50 apiece and sells them for \$1.00 apiece. If there are no other expenses, how many roses must be sold in order to make a profit of \$300 ?

   (A) 100
   (B) 150
   (C) 200
   (D) 300
   (E) 600

3. If $x \div 7 = 392 \div 8$, then $x =$

   (A) 343
   (B) 336
   (C) 329
   (D) 322
   (E) 315

4. The fraction $\frac{4}{3}$ is NOT between

   (A) $\frac{1}{2}$ and $\frac{3}{2}$

   (B) $\frac{2}{3}$ and 1

   (C) 1 and 2

   (D) $\frac{5}{4}$ and $\frac{6}{4}$

   (E) $\frac{6}{5}$ and $\frac{7}{5}$

5. If the average (arithmetic mean) of -5 and x is -5, then $x =$

   (A) 10
   (B) 5
   (C) 0
   (D) -5
   (E) -10

6. The perimeter of a 7-sided figure is 15. If the length of each side of the figure is increased by 2 units, what is the perimeter of the new figure?

   (A) 22 (B) 24 (C) 29 (D) 30 (E) 119

7. What is the sum of all the positive integer factors of 20 ?

   (A) 12
   (B) 21
   (C) 30
   (D) 33
   (E) 42

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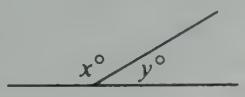
5

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Notes: 1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ●

	Column A	Column B
8.	The number of tens in 48	The number of hundreds in 348

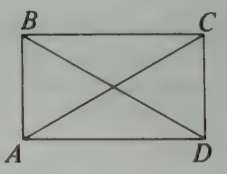
$x \neq 0$

9.	$\frac{1}{x^2}$	0
10.	$4(2x - 2)$	$2(4x - 4)$
11.	The length of the hypotenuse of a right triangle with legs of lengths 8 and 6	The length of the hypotenuse of a right triangle with legs of lengths 12 and 5

The sum of two positive numbers is 24.
The difference of the numbers is 10.

12.	The larger of these two numbers	18

	Column A	Column B
13.	$8 - (7 - 6 - 5)$	$8 - 7 - (6 - 5)$
14.	Volume of a rectangular block with dimensions 1 meter by 2 meters by 3 meters	Volume of a rectangular block with dimensions 1 meter by 12 meters by 0.5 meters



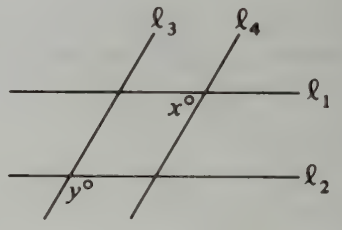
Note: Figure not drawn to scale.

In parallelogram $ABCD$, $\angle ABC \neq \angle BAD$.

15.	Length of AC	Length of BD

$2x + 2 < 1$

16.	x	0



Lines ℓ_1 and ℓ_2 are parallel.
Lines ℓ_3 and ℓ_4 are parallel.

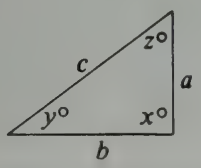
17.	x	$180 - y$

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SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

	Column A	Column B
	$x + y = 5$ $x - 2y = 7$	
18.	x	y
19.	Area of a circle with radius r	Area of a circle with diameter $2r$
	$x + y = y$	
20.	x	y



Note: Figure not drawn to scale.

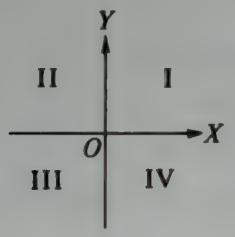
	$180 > x > y + z$	
21.	c^2	$a^2 + b^2$
	The average (arithmetic mean) of 18, 30, x, and y equals 12.	
	$x > 0$	
22.	y	0
	Juan is h years old and Lisa is k years older than Juan.	
23.	$k - h$	$h - k$
24.	Remainder when 731^{500} is divided by 10	1

	Column A	Column B
	x is a 2-digit number divisible by 2, 3, and 7.	
25.	x	84
	$m > 0$	
26.	$\frac{m}{0.25}$	$3m$
	$a^2 = b^2$	
27.	ab	a^2

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5

Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.



28. In the figure above, which quadrants contain pairs (x, y) that satisfy the condition $\frac{x}{y} = 1$?

(A) I only
(B) I and II only
(C) I and III only
(D) II and IV only
(E) I, II, III, and IV

---

29. If it is now 4:00 p.m. Saturday, in 253 hours from now, what time and day will it be? (Assume no daylight saving time changes in the period.)

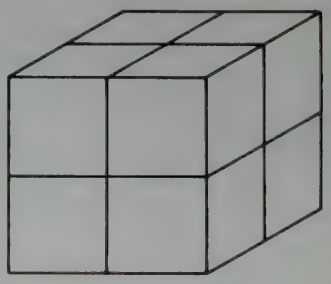
(A) 5:00 a.m. Saturday
(B) 1:00 a.m. Sunday
(C) 5:00 p.m. Tuesday
(D) 1:00 a.m. Wednesday
(E) 5:00 a.m. Wednesday

30. If $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ is defined to equal $ab - cd$ and $\begin{vmatrix} a & b \\ c & d \end{vmatrix} + y = 0$, then $y =$

(A) $ab - cd$
(B) $ac - bd$
(C) $ad - bc$
(D) $bc - ad$
(E) $cd - ab$

31. Milk costs x cents per half-gallon and y cents per gallon. If a gallon of milk costs z cents less than 2 half-gallons, which of the following equations must be true?

(A) $x - 2y + z = 0$
(B) $2x - y + z = 0$
(C) $x - y - z = 0$
(D) $2x - y - z = 0$
(E) $x + 2y - z = 0$



32. The large solid wooden cube above is composed of 8 smaller cubes of equal size and the total surface of the large cube is painted purple. If each of the smaller cubes is then cut into 8 cubes of equal size, how many of the smallest cubes will have exactly 3 purple faces?

(A) 4 (B) 8 (C) 16 (D) 24 (E) 32

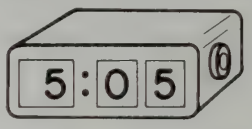
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33. In a 10-kilogram solution of water and alcohol, the ratio by mass of water to alcohol is 3:2. If 6 kilograms of a solution consisting of 2 parts water to 1 part alcohol is added to the 10-kilogram solution, what fraction by mass of the new solution is alcohol?

(A) $\frac{5}{8}$ (B) $\frac{3}{8}$ (C) $\frac{1}{3}$ (D) $\frac{5}{16}$ (E) $\frac{3}{16}$

34. If it takes 10 people 12 hours to do a certain job, how many hours would it take 6 people, working at the same rate, to do $\frac{1}{4}$ of the same job?

(A) 6 (B) 5 (C) $4\frac{1}{2}$ (D) 4 (E) $3\frac{3}{4}$



35. The twelve-hour digital clock above shows one example of a time at which the number representing the hour is equal to the number representing the minutes. What is the LEAST possible number of minutes from the instant one such "double" reading appears to the instant the next appears?

(A) 11 (B) 30 (C) 49 (D) 60 (E) 61

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

Correct Answers for Scholastic Aptitude Test
Form Code 2E

VERBAL		MATHEMATICAL	
Section 1	Section 3	Section 2	Section 5
1. B	1. B	1. D	1. A
2. A	2. C	2. D	2. E
3. E	3. B	3. A	3. A
4. E	4. E	4. E	4. B
5. C	5. D	5. A	5. D
6. D	6. C	6. C	6. C
7. B	7. E	7. A	7. E
8. B	8. C	8. E	*8. A
9. B	9. E	9. B	*9. A
10. D	10. E	10. B	*10. C
11. C	11. D	11. E	*11. B
12. C	12. A	12. B	*12. B
13. C	13. B	13. D	*13. A
14. B	14. D	14. D	*14. C
15. C	15. E	15. D	*15. D
16. C	16. A	16. E	*16. B
17. A	17. D	17. †	*17. C
18. B	18. E	18. C	*18. A
19. D	19. E	19. A	*19. C
20. C	20. D	20. B	*20. D
21. D	21. C	21. A	*21. A
22. E	22. A	22. A	*22. B
23. D	23. D	23. C	*23. D
24. A	24. C	24. E	*24. C
25. C	25. B	25. D	*25. D
26. E	26. B		*26. A
27. B	27. D		*27. D
28. A	28. C		28. C
29. B	29. D		29. E
30. C	30. E		30. E
31. D	31. C		31. D
32. D	32. C		32. B
33. A	33. A		33. B
34. D	34. E		34. B
35. D	35. D		35. C
36. B	36. D		
37. E	37. D		
38. E	38. E		
39. C	39. C		
40. B	40. D		
41. B			
42. B			
43. A			
44. B			
45. B			

*Indicates four-choice questions. (All of the other questions are five-choice.)

†Not scored

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

  $$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.
- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

You can verify the College Board SAT scores reported to you recently by using the information in this booklet along with the copy of your answer sheet. *Before you begin, check that the first two characters (number and letter) of the form code you marked in item 3 on your answer sheet are the same as the form code printed on the front of this booklet.* Compare the responses shown on the copy of your answer sheet with the list of correct answers.

SAT-Verbal Sections 1 and 3

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 3* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on page 174. Enter this figure on the worksheet.

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on page 174. Enter this figure on the worksheet.

SAT SCORING WORKSHEET

FORM CODE 2E

SAT-Verbal Sections

A. Section 1: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 3: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on the back cover.) [ ] SAT-verbal score

SAT-Mathematical Sections

A. Section 2: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____ no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on the back cover.) [ ] SAT-math score

Score Conversion Table
Scholastic Aptitude Test
Form Code 2E

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	460	620
84	790		39	450	610
83	780		38	450	600
82	770		37	440	590
81	760		36	430	580
80	750		35	430	570
79	740		34	420	560
78	730		33	410	550
77	720		32	410	540
76	710		31	400	540
75	700		30	390	530
74	690		29	390	520
73	680		28	380	510
72	670		27	380	500
71	670		26	370	490
70	660		25	360	480
69	650		24	360	470
68	640		23	350	460
67	630		22	340	460
66	630		21	330	450
65	620		20	330	440
64	610		19	320	430
63	610		18	310	420
62	600		17	310	410
61	590		16	300	400
60	580		15	290	390
59	580	800	14	290	390
58	570	790	13	280	380
57	560	780	12	270	370
56	560	770	11	260	360
55	550	760	10	260	350
54	550	750	9	250	340
53	540	740	8	240	330
52	530	730	7	230	320
51	530	720	6	230	310
50	520	710	5	220	300
49	510	700	4	210	300
48	510	690	3	200	290
47	500	690	2	200	280
46	500	680	1	200	270
45	490	670	0	200	260
44	480	660	– 1	200	250
43	480	650	– 2	200	240
42	470	640	– 3	200	230
41	470	630	– 4	200	220
			– 5	200	210
			– 6	200	210
			– 7 or below	200	200